Transformations

Name _____ Date _____ No calculator Version 1

Scan to practise this topic online

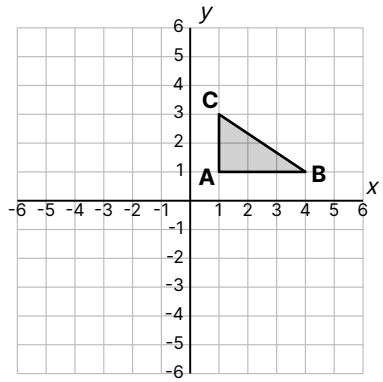
KEY WORDS translation *(nekehanga)* · reflection *(whakaata)* · rotation *(hurihanga)* · enlargement *(whakarahi)*

A Translation ●○○

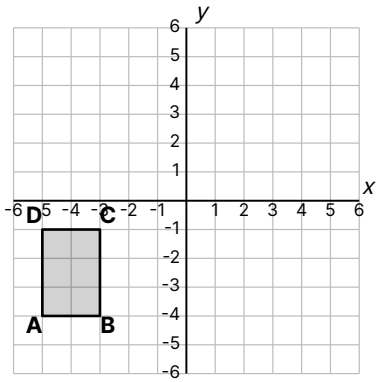
Remember: A translation slides every point the same distance in the same direction. Label the image A′, B′, C′.

1. Draw the image of each shape.

a) Translate the shape 5 units left and 2 units up.

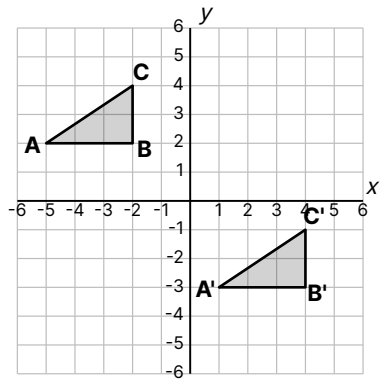


b) Translate the shape 7 units right and 3 units up.



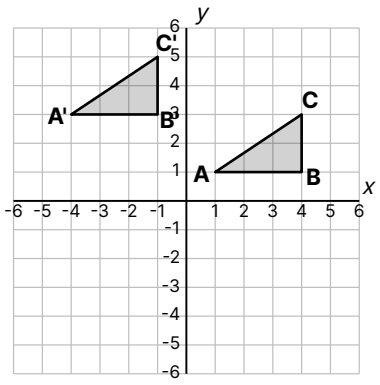
2. Each shape ABC has been translated to A′B′C′. Describe the translation.

a)



_____

b)



_____

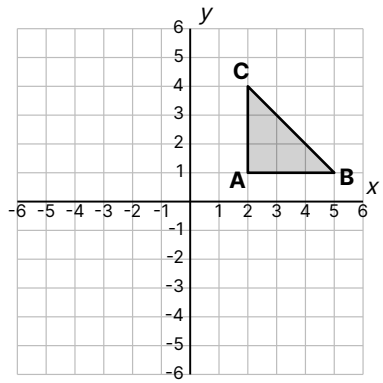
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B Reflection ●●○

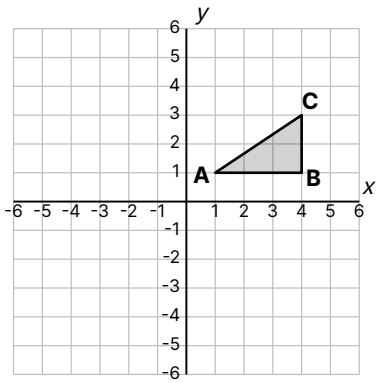
Remember: Each point and its image are the same distance from the mirror line, on opposite sides of it.

3. Draw the image of each shape.

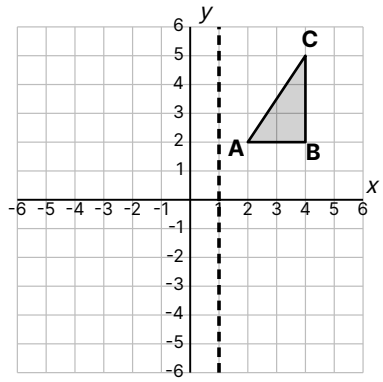
a) Reflect the shape in the y-axis.



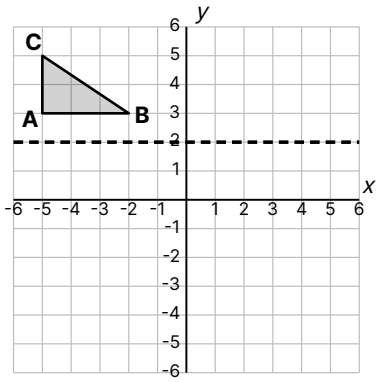
b) Reflect the shape in the x-axis.



c) Reflect the shape in the line $x = 1$ (the dashed line).



d) Reflect the shape in the line $y = 2$ (the dashed line).

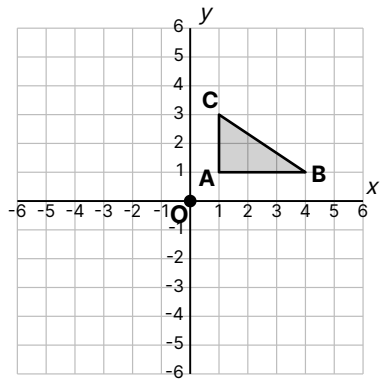


C Rotation ●●●

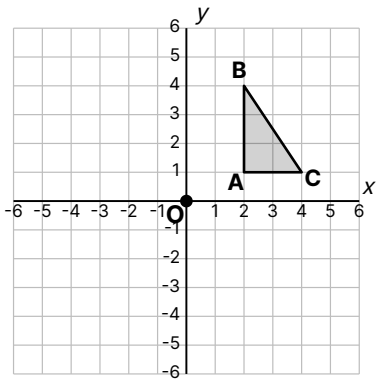
Remember: Every point turns through the same angle about the centre. Tracing paper helps: trace the shape, hold the centre still and turn.

4. Draw the image of each shape.

a) Rotate the shape 180° about the point O.

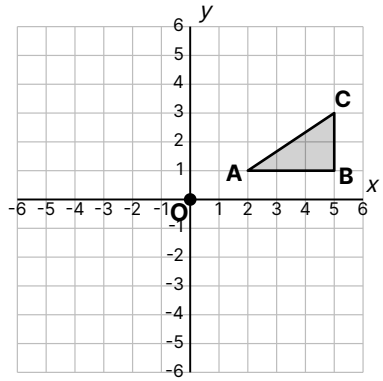


b) Rotate the shape 90° clockwise about the point O.

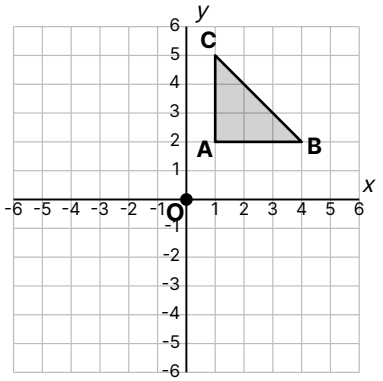


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c) Rotate the shape 90° anticlockwise about the point O.



d) Rotate the shape 90° clockwise about the point O.

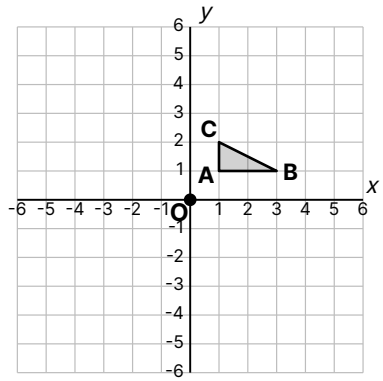


D Enlargement ●●●

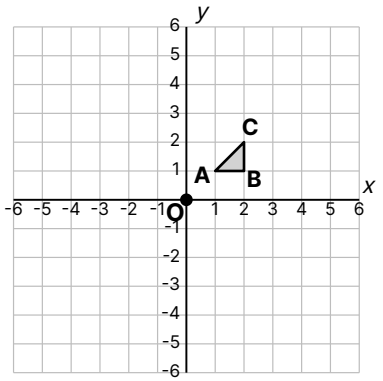
Remember: Multiply the distance from the centre to each point by the scale factor. A scale factor less than 1 makes the shape smaller.

5. Draw the image of each shape.

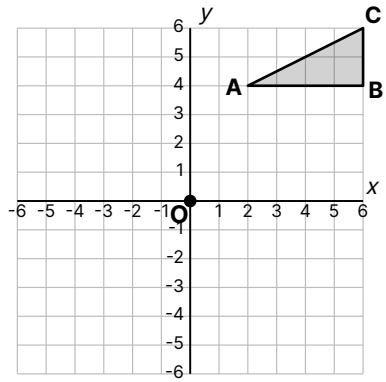
a) Enlarge the shape by a scale factor of 2, with centre O.



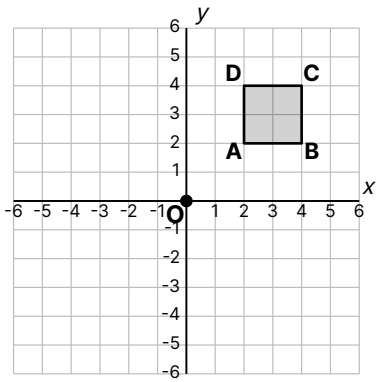
b) Enlarge the shape by a scale factor of 3, with centre O.



c) Enlarge the shape by a scale factor of 0.5, with centre O.



d) Enlarge the shape by a scale factor of 1.5, with centre O.



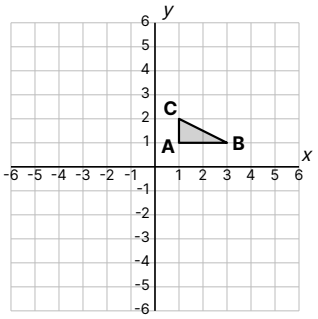
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E Combined transformations ●●●

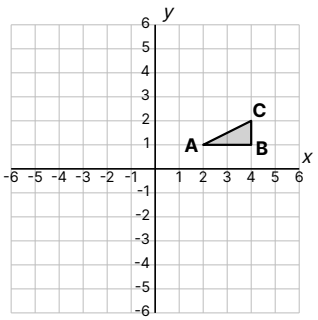
Remember: Do the first transformation, then do the second one to the new shape.

6. Draw the final image of each shape.

a) Translate the shape 2 units right and 3 units up. Then reflect the new shape in the y-axis.



b) Reflect the shape in the x-axis. Then translate the new shape 6 units left and 2 units down.

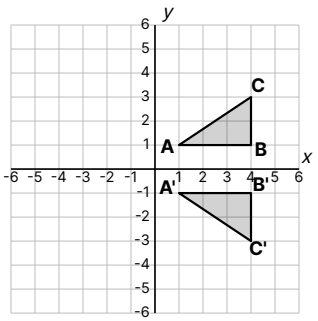


F Naming the transformation ●●●

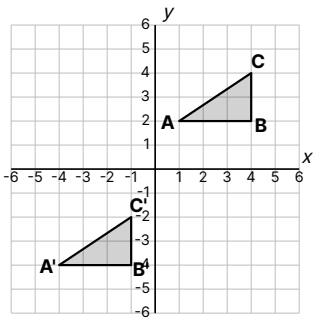
Remember: Translation: a slide. Reflection: a flip. Rotation: a turn. Enlargement: a change of size.

7. Shape ABC has been transformed to A′B′C′. Name and describe each transformation.

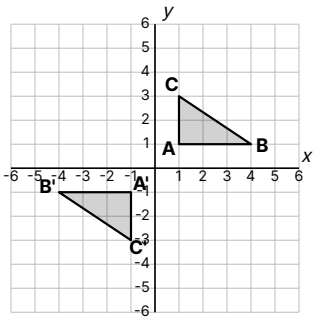
a)



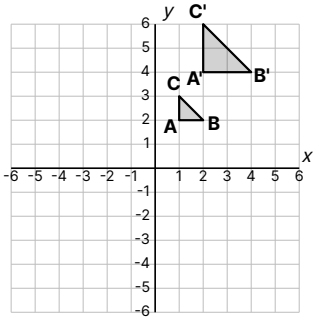
b)



c)



d)



★ Extension

8. Without drawing, write the coordinates of the image of the point $P(4,\ 2)$ after each transformation.

a) Translation 6 left and 1 up _____

b) Reflection in the x-axis _____

c) Reflection in the y-axis _____

d) Rotation of 180° about (0, 0) _____

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Transformations

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

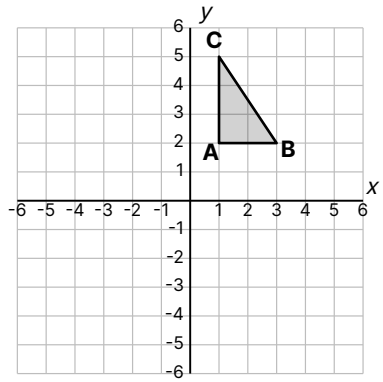
KEY WORDS translation *(nekehanga)* · reflection *(whakaata)* · rotation *(hurihanga)* · enlargement *(whakarahi)*

A Translation ●○○

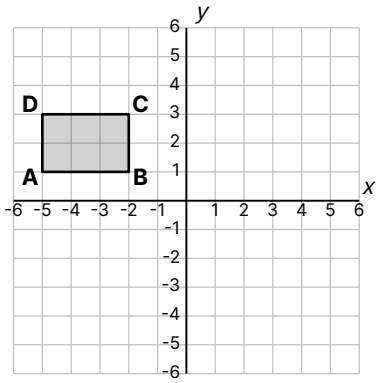
Remember: A translation slides every point the same distance in the same direction. Label the image A′, B′, C′.

1. Draw the image of each shape.

a) Translate the shape 6 units left and 4 units down.

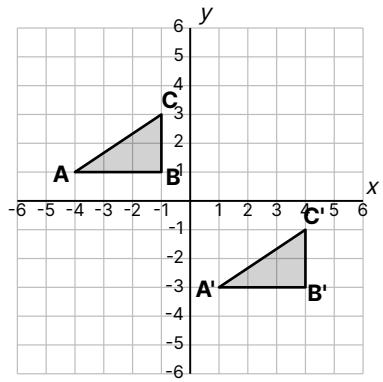


b) Translate the shape 6 units right and 5 units down.



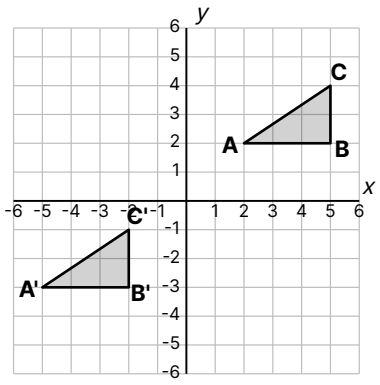
2. Each shape ABC has been translated to A′B′C′. Describe the translation.

a)



_____

b)



_____

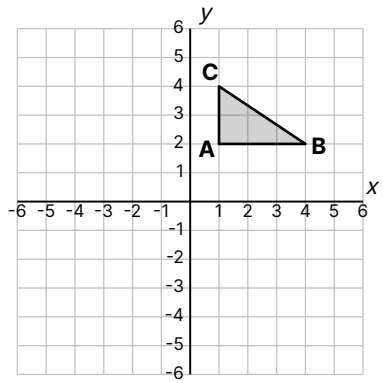
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B Reflection ●●○

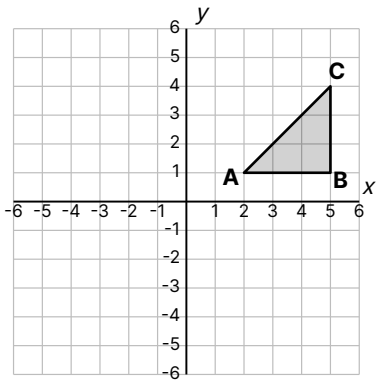
Remember: Each point and its image are the same distance from the mirror line, on opposite sides of it.

3. Draw the image of each shape.

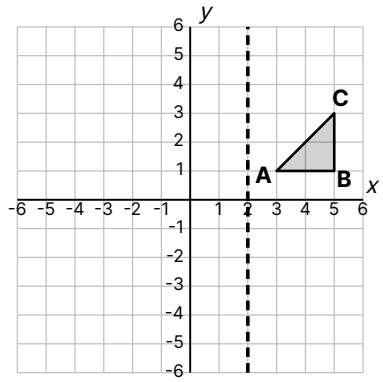
a) Reflect the shape in the y-axis.



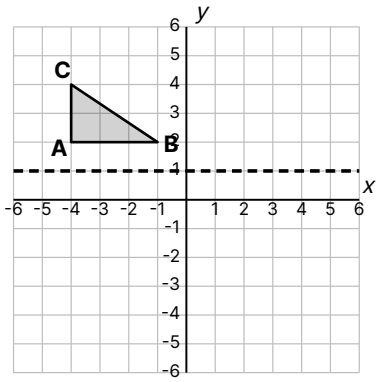
b) Reflect the shape in the x-axis.



c) Reflect the shape in the line $x = 2$ (the dashed line).



d) Reflect the shape in the line $y = 1$ (the dashed line).

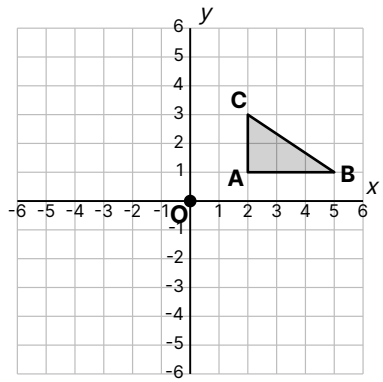


C Rotation ●●●

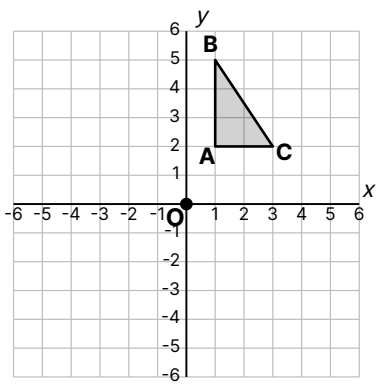
Remember: Every point turns through the same angle about the centre. Tracing paper helps: trace the shape, hold the centre still and turn.

4. Draw the image of each shape.

a) Rotate the shape 180° about the point O.

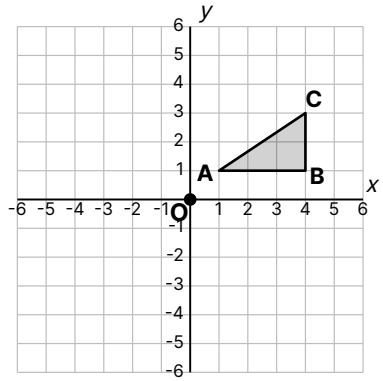


b) Rotate the shape 90° clockwise about the point O.

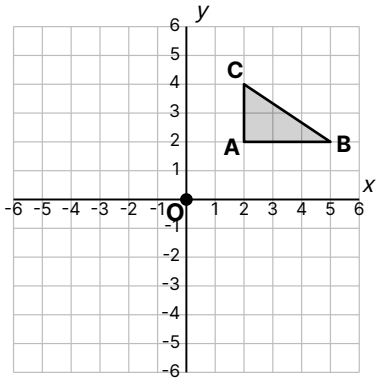


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c) Rotate the shape 90° anticlockwise about the point O.



d) Rotate the shape 90° clockwise about the point O.

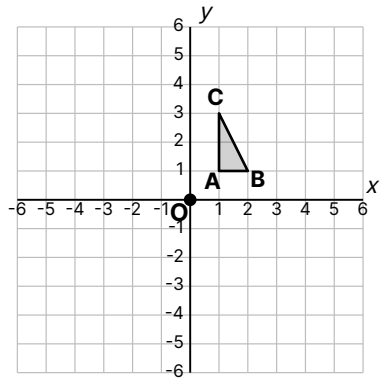


D Enlargement ●●●

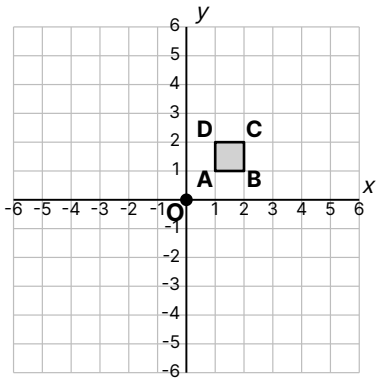
Remember: Multiply the distance from the centre to each point by the scale factor. A scale factor less than 1 makes the shape smaller.

5. Draw the image of each shape.

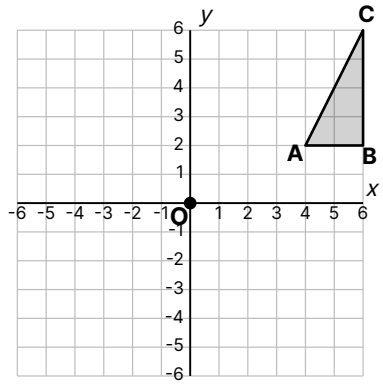
a) Enlarge the shape by a scale factor of 2, with centre O.



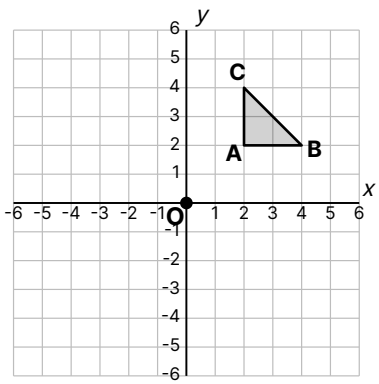
b) Enlarge the shape by a scale factor of 3, with centre O.



c) Enlarge the shape by a scale factor of 0.5, with centre O.



d) Enlarge the shape by a scale factor of 1.5, with centre O.



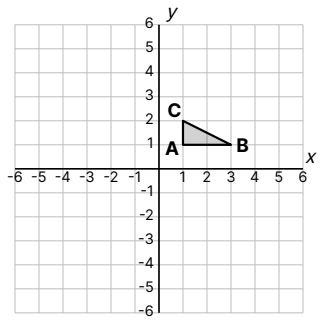
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E Combined transformations ●●●

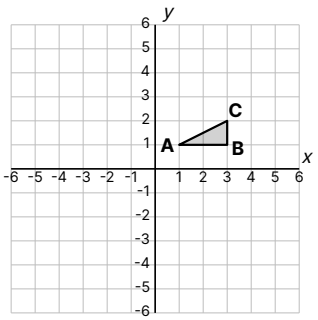
Remember: Do the first transformation, then do the second one to the new shape.

6. Draw the final image of each shape.

a) Translate the shape 3 units right and 2 units up. Then reflect the new shape in the y-axis.



b) Reflect the shape in the x-axis. Then translate the new shape 5 units left and 3 units down.

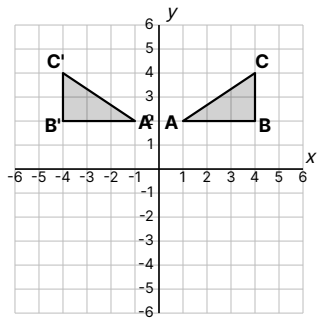


F Naming the transformation ●●●

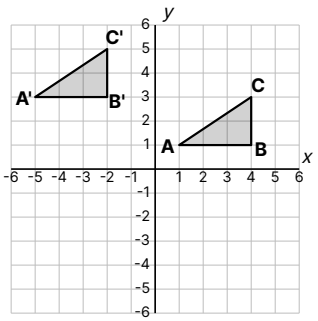
Remember: Translation: a slide. Reflection: a flip. Rotation: a turn. Enlargement: a change of size.

7. Shape ABC has been transformed to A′B′C′. Name and describe each transformation.

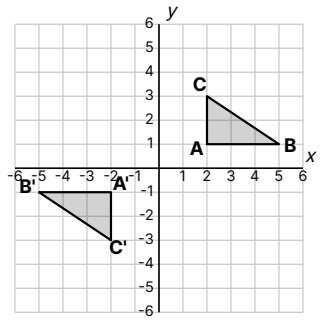
a)



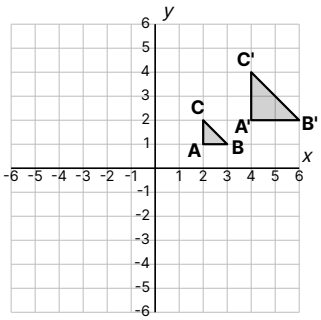
b)



c)



d)



★ Extension

8. Without drawing, write the coordinates of the image of the point $P(3,\ 5)$ after each transformation.

a) Translation 6 left and 1 up _____

b) Reflection in the x-axis _____

c) Reflection in the y-axis _____

d) Rotation of 180° about (0, 0) _____

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